1) a)
Stretch, parallel to the x-axis, scale factor 2

1) b)
Enlargement, centred at the origin, scale factor 7
Alternative Method:
Stretch, parallel to the x-axis, scale factor 7, and a Stretch, parallel to the y-axis, scale factor 7

1) c)
Stretch, parallel to the y-axis, scale factor 5

1) d)
Stretch, parallel to the x-axis, scale factor 10, and a Stretch, parallel to the y-axis, scale factor 20

1) e)
Enlargement, centred at the origin, scale factor $\frac{1}{3}$
Alternative Method:
Stretch, parallel to the x-axis, scale factor $\frac{1}{3}$, and a Stretch, parallel to the y-axis, scale factor $\frac{1}{3}$

2) a)
$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

2) b)
$\mathbf{A}^2 = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$

2) c)
$\mathbf{A}^3 = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$

2) d)
$\mathbf{A}^{15} = \begin{pmatrix} 32768 & 0 \\ 0 & 32768 \end{pmatrix}$

3)
The matrix represents a Stretch, parallel to the y-axis, scale factor 3. So the image will be $3 \times$ the area of the original shape. So the area of the shape's image is $3 \times 4 = 12$ units^2

4)
The matrix represents an Enlargement, centred at the origin, scale factor 3. So the image will be $3^2 \times$ the area of the original shape. So the area of the shape's image is $3^2 \times 4 = 36$ units^2

5)
$\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}\begin{pmatrix} -2 & 0 & 4 \\ 0 & 9 & 0 \end{pmatrix} = \begin{pmatrix} -8 & 0 & 16 \\ 0 & 36 & 0 \end{pmatrix}$
The coordinates of the vertices of triangle T' are $(-8, 0)$, $(0, 36)$ and $(16, 0)$
The area of the triangle is $= \frac{1}{2} \times 24 \times 36 = 432$ units^2

6) a)
$\begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}$

6) b)
$\begin{pmatrix} 15 & 0 \\ 0 & 15 \end{pmatrix}$

6) c)
$\begin{pmatrix} 1 & 0 \\ 0 & \frac{2}{3} \end{pmatrix}$

6) d)
$\begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$

7) a)
$\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ 2k \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$
$2k = 4 \Rightarrow k = 2$
$\therefore \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$

7) b)

$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 12 \\ -1 \end{pmatrix} \Rightarrow \begin{pmatrix} 4k \\ -1 \end{pmatrix} = \begin{pmatrix} 12 \\ -1 \end{pmatrix}$

$4k = 12 \Rightarrow k = 3$

$\therefore \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$

7) c)

$\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} 16 \\ 9 \end{pmatrix} = \begin{pmatrix} 16 \\ 3 \end{pmatrix} \Rightarrow \begin{pmatrix} 16 \\ 9k \end{pmatrix} = \begin{pmatrix} 16 \\ 3 \end{pmatrix}$

$9k = 3 \Rightarrow k = \frac{1}{3}$

$\therefore \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$

7) d)

$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 25 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \end{pmatrix} \Rightarrow \begin{pmatrix} 25k \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$

$25k = 5 \Rightarrow k = \frac{1}{5}$

$\therefore \begin{pmatrix} \frac{1}{5} & 0 \\ 0 & 1 \end{pmatrix}$

7) e)

$\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} 8 \\ -7 \end{pmatrix} = \begin{pmatrix} 8 \\ -3 \end{pmatrix} \Rightarrow \begin{pmatrix} 8 \\ -7k \end{pmatrix} = \begin{pmatrix} 8 \\ -3 \end{pmatrix}$

$-7k = -3 \Rightarrow k = \frac{3}{7}$

$\therefore \begin{pmatrix} 1 & 0 \\ 0 & \frac{3}{7} \end{pmatrix}$

7) f)

$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -30 \\ 5 \end{pmatrix} = \begin{pmatrix} -18 \\ 5 \end{pmatrix} \Rightarrow \begin{pmatrix} -30k \\ 5 \end{pmatrix} = \begin{pmatrix} -18 \\ 5 \end{pmatrix}$

$-30k = -18 \Rightarrow k = \frac{18}{30} = \frac{3}{5}$

$\therefore \begin{pmatrix} \frac{3}{5} & 0 \\ 0 & 1 \end{pmatrix}$

8) a)

$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 8 \\ 20 \end{pmatrix} \Rightarrow \begin{pmatrix} 2k \\ 5k \end{pmatrix} = \begin{pmatrix} 8 \\ 20 \end{pmatrix}$

$2k = 8 \Rightarrow k = 4$

$5k = 20 \Rightarrow k = 4$

$\therefore \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$

8) b)

$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} -240 \\ 90 \end{pmatrix} = \begin{pmatrix} -48 \\ 18 \end{pmatrix} \Rightarrow \begin{pmatrix} -240k \\ 90k \end{pmatrix} = \begin{pmatrix} -48 \\ 18 \end{pmatrix}$

$-240k = -48 \Rightarrow k = \frac{1}{5}$

$90k = 18 \Rightarrow k = \frac{1}{5}$

$\therefore \begin{pmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{5} \end{pmatrix}$

8) c)

$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} 20 \\ 600 \end{pmatrix} = \begin{pmatrix} 64 \\ 1920 \end{pmatrix} \Rightarrow \begin{pmatrix} 20k \\ 600k \end{pmatrix} = \begin{pmatrix} 64 \\ 1920 \end{pmatrix}$

$20k = 64 \Rightarrow k = \frac{16}{5}$

$600k = 1920 \Rightarrow k = \frac{16}{5}$

$\therefore \begin{pmatrix} \frac{16}{5} & 0 \\ 0 & \frac{16}{5} \end{pmatrix}$

8) d)

$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} -360 \\ -440 \end{pmatrix} = \begin{pmatrix} -342 \\ -418 \end{pmatrix} \Rightarrow \begin{pmatrix} -360k \\ -440k \end{pmatrix} = \begin{pmatrix} -342 \\ -418 \end{pmatrix}$

$-360k = -342 \Rightarrow k = \frac{19}{20}$

$-440k = -418 \Rightarrow k = \frac{19}{20}$

$\therefore \begin{pmatrix} \frac{19}{20} & 0 \\ 0 & \frac{19}{20} \end{pmatrix}$

9) a)

Area of $T = \frac{1}{2} \times (8-4) \times (9-3) = \frac{1}{2} \times 4 \times 6 = 12$ units^2

9) b)

$\begin{pmatrix} 5 & 0 \\ 0 & 12 \end{pmatrix} \begin{pmatrix} 4 & 8 & 7 \\ 3 & 3 & 9 \end{pmatrix} = \begin{pmatrix} 20 & 40 & 35 \\ 36 & 36 & 108 \end{pmatrix}$

So $A'(20, 36)$, $B'(40, 36)$ and $C'(35, 108)$

Area of $V = \frac{1}{2} \times (40-20) \times (108-36) = \frac{1}{2} \times 20 \times 72 = 720$ units^2

Alternative Method (using determinants):

$\begin{vmatrix} 5 & 0 \\ 0 & 12 \end{vmatrix} = 5 \times 12 - 0 \times 0 = 60$

Area of $V = 60 \times$ Area of $T = 60 \times 12 = 720$ units^2

10)

Area of $K = \frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 2 \times 5 = 7$

$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 & 1 \\ -5 & 0 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -a & 0 & a \\ -5a & 0 & 2a & 0 \end{pmatrix}$

Area of $K' = \frac{1}{2} \times 2a \times 2a + \frac{1}{2} \times 2a \times 5a = 7a^2$

$7a^2 = 2 \times 7 \Rightarrow a^2 = 2 \Rightarrow a = \sqrt{2}$

So K' has coordinates $(0, -5\sqrt{2})$, $(-\sqrt{2}, 0)$, $(0, 2\sqrt{2})$ and $(\sqrt{2}, 0)$

11) a)

$\begin{pmatrix} 7 & 0 \\ 0 & 3 \end{pmatrix}$

11) b)

$\begin{pmatrix} 7 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2m+n \\ n-m \end{pmatrix} = \begin{pmatrix} 18n+1 \\ m-1 \end{pmatrix} \Rightarrow \begin{pmatrix} 14m+7n \\ 3n-3m \end{pmatrix} = \begin{pmatrix} 18n+1 \\ m-1 \end{pmatrix}$

$14m + 7n = 18n + 1 \Rightarrow 14m - 11n = 1$ (1)

$3n - 3m = m - 1 \Rightarrow 4m - 3n = 1$ (2)

$\therefore m = 4$ and $n = 5$

12)

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \begin{pmatrix} -2 \\ 8 \end{pmatrix} - \begin{pmatrix} 2 \\ -6 \end{pmatrix} = \begin{pmatrix} -4 \\ 14 \end{pmatrix}$

$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} -4 \\ 14 \end{pmatrix} = \begin{pmatrix} -12 \\ 42 \end{pmatrix}$

$\overrightarrow{OA'} = \overrightarrow{OB} + \begin{pmatrix} -12 \\ 42 \end{pmatrix} = \begin{pmatrix} 2 \\ -6 \end{pmatrix} + \begin{pmatrix} -12 \\ 42 \end{pmatrix} = \begin{pmatrix} -10 \\ 36 \end{pmatrix}$

So $A'(-10, 36)$

13)

$\overrightarrow{BA'} = \overrightarrow{OA'} - \overrightarrow{OB} = \begin{pmatrix} 10 \\ -1 \end{pmatrix} - \begin{pmatrix} -8 \\ -3 \end{pmatrix} = \begin{pmatrix} 18 \\ 2 \end{pmatrix}$

$\begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix} \begin{pmatrix} 18 \\ 2 \end{pmatrix} = \begin{pmatrix} 180 \\ 20 \end{pmatrix}$

$\overrightarrow{OA} = \overrightarrow{OB} + \begin{pmatrix} 180 \\ 20 \end{pmatrix} = \begin{pmatrix} -8 \\ -3 \end{pmatrix} + \begin{pmatrix} 180 \\ 20 \end{pmatrix} = \begin{pmatrix} 172 \\ 17 \end{pmatrix}$

So $A(172, 17)$